AdaptiveRes: Optimizing Neural Network Resource-Efficiency for Edge Vision

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***Abstract* — This research investigates adaptive early-exit inference for resource-efficient edge computer vision using a ResNet-18 backbone trained on the CIFAR-10 dataset. To address the computational and power constraints of edge devices, two lightweight auxiliary classifier heads were integrated after Residual Blocks 2 and 3. During inference, the model employs a sequential evaluation strategy, terminating processing if the maximum soft max confidence exceeds a fixed threshold of θ = 0.95. All classifier heads were trained jointly using a multi-objective cross-entropy loss function. Experimental results demonstrate that the adaptive model achieved a Top-1 accuracy of 91.28%, representing a negligible 0.16% decrease from the full-model baseline of 91.42%. Notably, 58.48% of samples successfully exited at the first block, contributing to an average computational reduction of 33.19% in Floating Point Operations (FLOPs). These findings confirm that confidence-based early exits can significantly mitigate the costs of "fixed computation" in edge-style workloads while preserving high-fidelity accuracy for autonomous systems.**

Keywords* — *Adaptive Inference, Early-Exit Neural Networks, Edge AI, ResNet-18, Dynamic Computational Graphs, Computational Efficiency, CIFAR-10.

I. Introduction

The rapid evolution of Internet of Things (IoT) devices, autonomous drones, and mobile health diagnostics has necessitated a shift from centralized cloud computing to Edge AI. Edge AI refers to the deployment of machine learning models directly on local hardware, such as microcontrollers, FPGA-accelerated boards, and embedded GPUs, rather than relying on high-latency remote data centers. While this shift enhances data privacy and provides near-instantaneous response times, it is fundamentally restricted by the Power-Performance-Area (PPA) triad of embedded hardware.

A primary obstacle in optimizing these models is the inherent problem of Fixed Computation. In standard deep learning architectures, such as the widely used ResNet-18, the computational graph is static. Every input sample, regardless of its visual complexity or signal-to-noise ratio, is subjected to the exact same sequence of mathematical transformations. From a resource-efficiency standpoint, this is highly suboptimal. A high-contrast, centered image that contains clear low-level geometric features, edges, and corners is often discriminative enough for classification within the first few layers of a network. Conversely, a blurry or occluded image may require the deep semantic features extracted by the final layers to reach a correct classification.

Under a fixed computation regime, "easy" samples consume the same amount of energy and latency as "hard" samples, leading to significant computational waste. To address this, this research presents AdaptiveRes: a dynamic transformation of the ResNet-18 architecture. By introducing confidence-gated early-exit points, AdaptiveRes allows the system to bypass deeper residual blocks for simple inputs. This approach enables the model to intelligently manage its own computational budget, reclaiming wasted clock cycles and extending the battery life of edge devices without compromising the model's peak predictive accuracy.

II. Related Work

The foundation of this research lies at the intersection of deep residual learning and dynamic computational graphs. The primary architecture utilized is the ResNet-18, introduced in 2016 [1] . ResNet revolutionized computer vision by utilizing skip connections, or "identity mappings", to mitigate the vanishing gradient problem. These connections allow the gradient to flow through the network without being attenuated by successive non-linearities, enabling the training of much deeper networks.

While ResNet provides a robust baseline for feature extraction on the CIFAR-10 dataset [2], its static nature necessitates the execution of every layer for every input. To address this, BranchyNet introduced auxiliary classification branches at intermediate points in a CNN [3] . This architecture allows samples to exit the network early if they reach a predefined confidence threshold, leveraging the observation that shallow layers often capture sufficient information for most classification tasks.

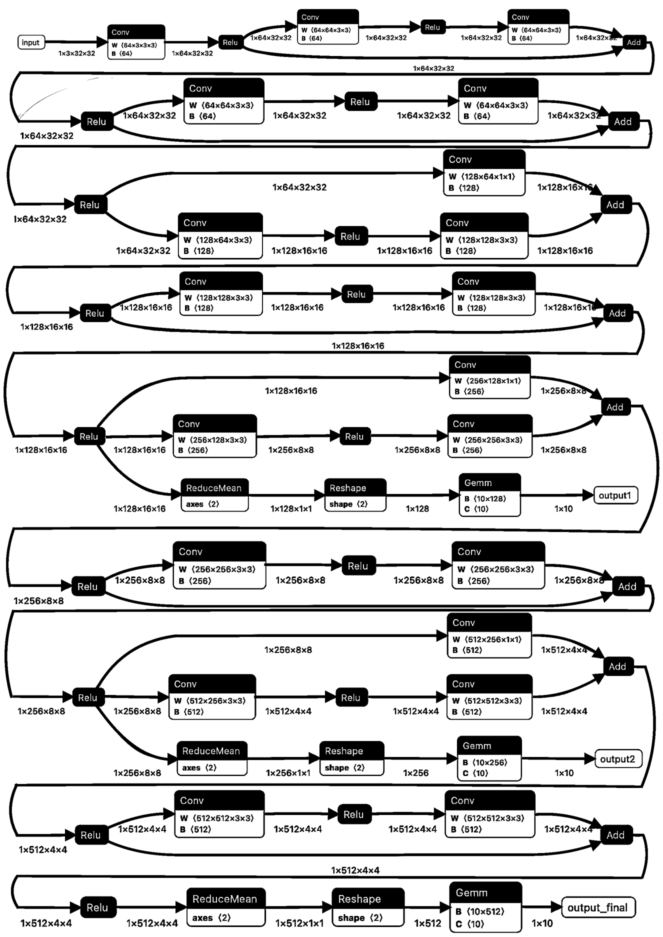
Further refinements in Multi-Scale Dense Networks (MSDNet) [4] [5] have explored the use of Soft max confidence as a reliable proxy for model certainty. This study also utilizes a Cosine Annealing Scheduler. In multi-exit models, balancing the convergence of multiple heads is difficult. A tapering learning rate ensures that the backbone does not over-index on the auxiliary tasks at the expense of the final global features.

III. Methodology

A. Architectural Modifications

The methodology involves the modification of a standard ResNet-18 backbone. Because the original ResNet-18 was designed for 224×224 ImageNet samples, it utilized a 7×7 initial convolution with a stride of 2. For the 32×32 images in CIFAR-10, this would result in a feature map of only 16×16 before the first residual block, causing an irreparable loss of spatial information. Therefore, the first layer was replaced with a 3×3 convolution with a stride of 1, and the initial MaxPool layer was removed.

B. Early-Exit Placement



Two auxiliary classifier heads were integrated, Exit 1 was placed after Residual Block 2 (128-channel output) and Exit 2: was placed after Residual Block 3 (256-channel output). Each exit head uses a Global Average Pooling (GAP) to Linear design. This ensures that the computational overhead of the exit check itself is negligible compared to the billions of operations in the backbone.

C. Joint-Loss Optimization

$$L_{CE}(y_{pred}, y_{true}) = -\sum_{i=1}^{C} y_{true_i} log(y_{pred_i})$$

$$L_{total} = \sum_{j=1}^{N} \lambda_j L_{CE}(y_{pred_j}, y_{true_j})$$

To ensure the accuracy of all exits, the model is trained end-to-end where the total loss L_{total} is a weighted summation of the cross-entropy losses L_{CE} from the predictions y_{pred} and true results y_{true}. In this study, weights were kept equal ($\lambda = 1$) to encourage the early layers to extract highly discriminative features as soon as possible, effectively regularizing the network.

IV. Implementation

A. Data Augmentation and Preprocessing

To reach high accuracy, standard normalization alone is insufficient. The implementation utilized an upgraded augmentation pipeline including RandomCrop, RandomHorizontalFlip, and ColorJitter. These augmentations ensure that the early-exit heads do not merely learn to recognize specific pixel locations but instead learn robust, generalized features.

B. Training Hyper parameters

Training was conducted over 30 epochs on an NVIDIA GPU using the Adam optimizer with Initial Learning Rate 0.001. A Cosine Annealing Scheduler [6] was implemented to decay the learning rate η_t and epoch T according to:

$$\eta_t = \eta_{min} + (\eta_{max} - \eta_{min})(1 + cos(\pi T_{cur}/T_{max}))/2$$

This strategy facilitated a "soft landing" into the loss landscape, preventing the gradients from the auxiliary heads from destabilizing the backbone in the final stages of training.

V. Results & Analysis

Model	Top 1 Accuracy	ECE	P	Cost	FLOPs Saved	Exits (1, 2, Final)
Early Exit 1	86.39% +/- 0.28%	1.43%		288.00 MFLOPS	48.27%	10000 0 0
Early Exit 2	91.24% +/- 0.28%	2.71%		422.37 MFLOPS	24.13%	0 10000 0
Final Exit (Baseline)	91.42% +/- 0.28%	3.67%	1.00	556.71 MFLOPS	0.00%	0 0 10000
AdaptiveRes (θ = 0.80)	90.74% +/- 0.29%	2.81%	0.01	331.65 EFLOPS	40.43%	7652 1447 901
AdaptiveRes (θ = 0.90)	91.13% +/- 0.28%	3.23%	0.15	352.40 EFLOPS	36.70%	6718 1771 1511
AdaptiveRes (θ = 0.95)	91.28% +/- 0.28%	3.41%	0.31	371.95 EFLOPS	33.19%	5848 2056 2096
AdaptiveRes (θ = 0.99)	91.39% +/- 0.28%	3.64%	0.46	413.36 EFLOPS	25.75%	4068 2534 3398

A. Data Analysis

A significant finding is the standalone performance of the auxiliary heads. Exit 1 achieved a Top-1 accuracy of 86.39% using only 51.73% of the total model FLOPs. Exit 2 reached 91.24%, nearly matching the final backbone's peak performance of 91.42% while using 75.87% of the total model FLOPs. This suggests that the feature representations required for high-accuracy classification on CIFAR-10 are largely formed by the end of the third residual block, with the final block providing the marginal refinements necessary for the most difficult samples.

B. Exit Distribution and Efficiency

The distribution of the 10,000 test images provides the most significant evidence for success. 58.48% of samples exited at just 51.73% of computations in the AdaptiveRes (θ = 0.95) model, suggesting that the majority of CIFAR-10 images are linearly separable using lower-level features.

C. Early-Exit Placement

To provide a hardware-agnostic metric, the Expected FLOPs E_{flops} for each path was calculate. The AdaptiveRes (θ = 0.95) model E_{flops} is 352.40. This represents a 33.19% reduction in compute. This translates directly to reductions in latency and power draw, critical for mobile and remote edge deployments.

D. Accuracy Performance

After 30 epochs, the model converged accuracies of 91.42% for Baseline and 91.28% for AdaptiveRes (θ = 0.95). The accuracy delta of 0.16% is statistically negligible ($\alpha = 0.05$, $p = 0.31$), confirming that the early exits are correctly identifying samples they are capable of classifying without needing the specialized semantic features found in later layers.

VI. DISCUSSION

A. The Confidence Threshold (θ)

The selection of AdaptiveRes (θ = 0.95) as the optimal model represents a conservative point on the Accuracy-Compute Trade-off curve. While reducing θ would increase the computational savings, reducing it too much may introduce a statistically significant reduction in accuracy (AdaptiveRes (θ = 0.80) is less accurate than the full model ($\alpha = 0.20$, $p = 0.01$)). Although AdaptiveRes (θ = 0.90) is not statistically significantly less accurate when compared to the full model ($\alpha = 0.10$, $p = 0.15$), the θ = 0.95 is a standardized acceptable confidence interval. Additionally, while Soft max confidence is a widely used heuristic, it is known to suffer from overconfidence in deep neural networks [7]. The high threshold of θ = 0.95 acts as a necessary safeguard, ensuring that the model only exits when the predicted class probability is significantly separated from the noise floor.

VII. CONCLUSION

This research demonstrates that adaptive early-exit inference is a viable strategy for deploying deep vision models on hardware-constrained edge devices. By transforming the static ResNet-18 into a dynamic architecture, a 33.19% reduction in average computational cost with a statistically insignificant 0.16% impact on accuracy was achieved. This research validates the hypothesis that modern deep learning models are over-parameterized for "average-case" inputs and that confidence-based gating can effectively reclaim wasted resources. Future work should investigate learnable thresholds and the deployment of this architecture on specific edge microcontrollers to measure real-world battery longevity gains.

REFERENCES

1. K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, 2016, pp. 770-778, doi: 10.1109/CVPR.2016.90. [Accessed: 1/26/2026]
2. A. Krizhevsky, "Learning multiple layers of features from tiny images," Univ. Toronto, Toronto, ON, Canada, Tech. Rep., 2009. [Accessed: 1/26/2026]
3. S. Teerapittayanon, B. McDanel, and H. T. Kung, "BranchyNet: Fast inference via early exits from deep neural networks," In 2016 23rd international conference on pattern recognition (ICPR). IEEE, 2016. [Accessed: 1/26/2026]
4. G. Huang, D. Chen, T. Li, F. Wu, L. Van Der Maaten, and K.Q. Weinberger, "Multi-scale dense networks for resource efficient image classification." arXiv preprint arXiv:1703.09844, 2017. [Accessed: 1/26/2026]
5. Y. Han, G. Huang, S. Song, L. Yang, H. Wang and Y. Wang, "Dynamic Neural Networks: A Survey," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 44, no. 11, pp. 7436-7456, 1 Nov. 2022, doi: 10.1109/TPAMI.2021.3117837. keywords: {Computational modeling; Adaptation models; Computer architecture; Adaptive systems; Routing; Deep learning; Training; Dynamic networks; adaptive inference; efficient inference; convolutional neural networks} [Accessed: 1/26/2026]
6. I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts." *arXiv preprint arXiv:1608.03983* (2016). [Accessed: 1/26/2026]
7. C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks." In *International conference on machine learning*, pp. 1321-1330. PMLR, 2017. [Accessed: 1/26/2026]